PROJECT BRIEF · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 9: Project: Secret Messenger

Module 3 | 50 minutes | Project: cryptography app | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Your mission
Build a messenger that encrypts and decrypts with a Caesar cipher: a menu, message and key inputs, one shift procedure used for both directions, validation, and round-trip tests.

Checklist

- [ ] Menu
- [ ] Message and key inputs
- [ ] Shift procedure
- [ ] Wrap-around
- [ ] Validation
- [ ] Round-trip tests

Milestones

1. Menu and inputs — done on: _____
2. Shift procedure — done on: _____
3. Decrypt — done on: _____
4. Validation — done on: _____
5. Tested and submitted — done on: _____

Plan it

Sketch your stage and write what each sprite will do.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Cipher	Incorrect	Encrypts only	Encrypts and decrypts with wrap-around	Handles spaces, capitals and punctuation
Procedures	None	Separate copies	One shift procedure for both	Clean, reusable design
Robustness	Breaks	Some checks	Validated menu and key	Friendly handling of all inputs
Testing	None	Ran once	Three round-trip tests	Systematic tests including edge letters
Security reflection	None	Vague	Explains weaknesses	Proposes and justifies an improvement

Want more?

- Vigenère cipher
- Brute-force cracker
- Keep capitals

Two stars and a wish

Ask a friend to test your project.

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